SEP
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Criteria A and B
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Summative Summative 17/17 Students Tuesday at 11:35 AM
IB MYP Design - Ms Ishita (MYP2) B

Task Details

SA Template

ATL Skills (Category, Cluster, Indicator):

Research Skills – Cluster VI: Information Literacy
Collect, record, and verify data from primary and secondary sources to identify and provide evidence for a real-world problem.

Thinking Skills – Cluster VIII: Critical Thinking
Evaluate evidence, arguments, and existing products to justify a proposed design solution.

Concepts:
Key Concept: Communities
Related Concepts: Function, Innovation

Content/Topic:
Investigating a real community problem and researching, specifying, and planning a sensor-based automated system (sensor → microcontroller → actuator) that solves it within the global context of Scientific and Technical Innovation and ready to prototype in Wokwi.

Statement of Inquiry:
Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation:
You are a student design engineer completing the IB MYP Design Student Workbook for Sensing Our World. Communities everywhere face problems : safety risks, environmental hazards, accessibility gaps that go unnoticed or unresolved until someone senses them and responds. Your task is to identify a real community problem, research it thoroughly, analyse existing sensor-based products that address similar problems, and use your findings to design, specify, and plan an automated system, built from sensors (e.g. PIR, rain, gas, temperature, ultrasonic), an Arduino microcontroller, and actuators (e.g. LED, servo motor, buzzer, LCD, relay) that could sense the problem and trigger a response. Your ideas must be tested against clear specifications and justified with a decision matrix before you plan the system architecture and flow diagrams needed to prototype your solution in Wokwi.

Role:
A junior design engineer on a project investigating a real community problem and developing an evidence-based, feasible automated sensing system to solve it.

Audience:
The members of the community affected by your chosen problem (e.g. residents, the school community, or local stakeholders) who would use or benefit from your proposed system, and the client/design panel evaluating whether your solution is well-

researched and feasible.

Product:
A completed IB MYP Design Student Workbook: a Problem Statement with context, evidence, and global connections; a prioritized Research Plan (People, Planet, System); an Existing Product Analysis of three or more products; a Research Synthesis and Design Brief; a Design Specifications Document; three annotated Design Ideas; a Decision Matrix with a justified Final Choice; and a System Architecture Diagram and System Flow Diagram, ready to prototype in Wokwi.

Standards:
Your work will be assessed against MYP Design Criterion A (i–iv) and Criterion B (i–iv), detailed in the Task-Specific Clarifications below.

Attaching the ***Task Specific Clarification***

Criterion A: Inquiring and Analysing

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **states** the need for a solution to a problem ii. **states** the findings of research.	The student: i. **states** the community problem in one short paragraph (e.g. "people forget to bring water bottles"), with minimal to no information on context/effects/global connection, without stating why a solution is needed. ii. **states** two facts about the problem from research (primary/ secondary), without connecting them to the stated problem.
3-4	The student: i. **outlines** the need for a solution to a problem ii. **states** some points of research needed to develop a solution, with some guidance iii. **states** the main features of an existing product that inspires a solution to the problem iv. **outlines** some of the main findings of research.	The student: i. **outlines** the community problem, briefly noting where it happens or who it affects, along with context and how it connects globally, and provides the need for a solution. ii. **states** 3-4 basic questions across the Research Plan's People, Planet, and System branches and finds answers with some guidance, without prioritizing the questions. iii. **states** two-three main features of an existing product, without describing how it works. iv. **outlines** some findings from research, without clearly summarizing, leaving gaps or unclear links to the stated problem.
5-6	The student: i. **explains** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed to develop a solution, with some guidance iii. **outlines** the main features of an existing product that inspires a solution iv. **outlines** the main findings of relevant	The student: i. **explains** the community problem clearly on the Problem Statement and "Why Does This Problem Need a Solution?" cards, including at least one consequence of leaving it unsolved. ii. **states and prioritizes** research questions across the People, Planet, and System branches of the Research Plan, with some guidance. iii. **outlines** the main features of at least 3 existing products using an analysis framework; outlines the research synthesis, which

Achievement Level	Level Descriptors	Task Specific Clarifications
	research.	does not yet include the features to be adopted, adapted and avoided in their own solution. iv. **outlines** several relevant findings, drawing on both the Research Findings box and the Existing Product Analysis box (at least two products considered), with some explanation of how these findings relate to the needs of the user/client.
7-8	The student: i. **explains and justifies** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed, with minimal guidance iii. **describes** the main features of an existing product that inspires a solution iv. **presents** the main findings of relevant research.	The student: i. **explains and justifies** the community problem in detail across the Problem Statement, Context, and "Why Does This Problem Need a Solution?" cards, using evidence to show why a solution is needed. ii. **states and prioritizes** research questions across all three Research Plan branches independently, with minimal guidance, showing a clear rationale for what to investigate first. iii. **describes** in detail the main features (SWOT) of 3 or more existing products/systems on the Existing Product Analysis; states the research synthesis, which includes the features to be adopted, adapted, and avoided in their own solution. iv. **presents** the main findings clearly and in an organized way across the Research Findings and Existing Product Analysis boxes, explicitly connecting them to the user's needs, and uses these findings to justify the proposed solution concept and the design brief that follows.

Criterion B: Developing Ideas

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **lists** a few basic success criteria for the design of a solution ii. **presents** one design idea, which can be interpreted by others iii. **creates** incomplete planning drawings/diagrams.	The student: i. **lists** 1-2 basic points in the Design Specifications Document, with minimal to no connection to the survey/research data. ii. **presents** only one of the three Design Idea boxes, with a basic sketch or label and little to no annotation. iii. creates the System Architecture and/or Flowchart diagram with missing labels, arrows, or key components (Sensor, Arduino, Output).
3-4	The student: i. **constructs** a list of the success criteria for the design of a solution ii. **presents** a few feasible design ideas,	The student: i. **constructs** a short list of success criteria in the Design Specifications Document, with a loose or unclear link back to the Criterion A research/survey findings.

Achievement Level	Level Descriptors	Task Specific Clarifications
	using an appropriate medium(s) or explains key features, which can be interpreted by others iii. **outlines** the main reasons for choosing the design with reference to the design specification iv. **creates** planning drawings/diagrams or lists requirements for the chosen solution.	ii. **presents** 2-3 of the Design Idea boxes, with either a sketch or a brief written explanation of the concept, not both. iii. **outlines** an idea in "My Final Choice" and gives 1-2 sentences of reasoning that briefly mentions the specification, without matching each point. iv. **creates** one of the two planning diagrams (System Architecture or Flowchart) accurately, or lists some build requirements without a full diagram.
5-6	The student: i. **develops** design specifications, which identify the success criteria for the design of a solution. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and explains key features, which can be interpreted by others. iii. **presents** the chosen design and outlines the main reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and lists requirements for the creation of the chosen solution.	The student: i. **develops** a full Design Specifications Document with clear, identifiable success criteria drawn from the Criterion A research/ survey data. ii. **presents** all three Design Idea boxes, each with an annotated sketch/screenshot and a written explanation of the concept's key features. iii. **presents** a decision matrix to compare the three ideas, then outlines the main reasons for the final choice in "My Final Choice," referencing specific specification points. iv. **creates** both the System Architecture diagram (Sensor → Arduino → Output) and the Flowchart accurately, with labelled components and a basic list of requirements for the build.
7-8	The student: i. **develops** a design specification which outlines the success criteria for the design of a solution based on the data collected. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and annotation, which can be correctly interpreted by others. iii. **presents** the chosen design and outlines the reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and outlines requirements for the creation of the chosen solution.	The student: i. **develops** a precise Design Specifications Document where each success criterion is explicitly traceable to a specific piece of Criterion A data (e.g. survey figures, source findings). ii. **presents** fully annotates all three Design Idea boxes so a peer could interpret and understand each concept without further explanation, using clear sketches/screenshots and detailed feature notes. iii. **presents** the final choice with a detailed, point-by-point justification in "My Final Choice," explicitly linking each reason back to the decision matrix and design specification. iv. **develops** accurate, fully labelled System Architecture and Flowchart diagrams and outlines the complete sequence of requirements needed to build the prototype.

Assessment Criteria

A: Inquiring and analysing

Achievement Level	Descriptor
0	The student does not reach a standard described by any of the descriptors below.
1 - 2	The student: i. states the need for a solution to a problem ii. states some of the main findings of relevant research.
3 - 4	The student: i. outlines the need for a solution to a problem ii. states the research needed to develop a solution to the problem, with some guidance iii. outlines one existing product that inspires a solution to the problem iv. develops a basic design brief, which outlines some of the findings of relevant research.
5 - 6	The student: i. explains the need for a solution to a problem ii. constructs a research plan, which states and prioritizes the primary and secondary research needed to develop a solution to the problem, with some guidance iii. describes a group of similar products that inspire a solution to the problem iv. develops a design brief, which outlines the findings of relevant research.
7 - 8	The student: i. explains and justifies the need for a solution to a problem ii. constructs a research plan, which states and prioritizes the primary and secondary research needed to develop a solution to the problem independently iii. analyses a group of similar products that inspire a solution to the problem iv. develops a design brief, which presents the analysis of relevant research.

B: Developing ideas

Achievement Level	Descriptor
0	The student does not reach a standard described by any of the descriptors below.
1 - 2	The student: i. lists a few basic success criteria for the design of a solution ii. presents one design idea, which can be interpreted by others iii. creates incomplete planning drawings/diagrams.
3 - 4	The student: i. constructs a list of the success criteria for the design of a solution ii. presents a few feasible design ideas, using an appropriate medium(s) or explains key features, which can be interpreted by others iii. outlines the main reasons for choosing the design with reference to the design specification iv. creates planning drawings/diagrams or lists requirements for the chosen solution.

Achievement Level	Descriptor
5 - 6	The student: i. develops design specifications, which identify the success criteria for the design of a solution ii. presents a range of feasible design ideas, using an appropriate medium(s) and explains key features, which can be interpreted by others iii. presents the chosen design and outlines the main reasons for its selection with reference to the design specification iv. develops accurate planning drawings/diagrams and lists requirements for the creation of the chosen solution.
7 - 8	The student: i. develops a design specification which outlines the success criteria for the design of a solution based on the data collected ii. presents a range of feasible design ideas, using an appropriate medium(s) and annotation, which can be correctly interpreted by others iii. presents the chosen design and outlines the reasons for its selection with reference to the design specification iv. develops accurate planning drawings/diagrams and outlines requirements for the creation of the chosen solution.